

PAPER_1364_QUANTUM_BIOLOGY

Daniel T. Murphy

PAPER_1364 — Quantum Biology Coherence in Photosynthesis

Framework: UQFF — Star-Magic v5.27+ | Tier: G Bio/Complex (CLOSED)

Calculator: calculate_paradox({"paradox": "quantum_biology"})

Master Expression $T_{\text{coherence}} = \hbar \cdot \omega_{\text{SCm}} / k_B / \beta_i = 99.5 \text{ K}$

****Match: 29% from FMO ~77 K****

Interpretation SCm phonon-assisted exciton transport. FMO complex coherence at SCm phonon $\times \beta_i$ temperature scale.

UQFF Locked Primitives - $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$ - $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT UQFF derives via integer-primitive identity. Zero fit parameters.

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, Youngstown, OH.